Statistics

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Statistics

Introduction

Information (in form of facts or numbers) collected to be examined and considered and used to help in decision-making is called data.
e.g. People residing in society, salary of emloyee working in a company or score made by cricket team in first innings
Such data can be use for future planning and decision making.

Statistics

Science dealing with the collection, presentation, analysis and interpretation of numerical data.

Facts about data

(i) Qualitative Characteristics like beauty, honesty etc. can not be considered as data. Data should be Numerical

(ii) Atleast two observations should be there to form data.

(iii) Single data collected can be used in various ways

Data can be divided in two categories, Primary & Secondary, which depends upon whether it has been collected by the person himself or data being used is collected by some other party

Variable

A quantity capable of taking different values
Ex : Marks of candidate in exam is variable.

Variables can be divided into two types, first is
Continuous variable - Continuous variable is defined as a variable that can have any value in a certain range.
for example : weight of persons in a gym
Second is
Discrete variable - this is a variable which not necessarily takes all possible values between two given values
i.e. Number of person in a group can not have all values between 7 to 11, it can be 8, 9 but it cannot be 8.1

TERMS IN STATISTICS

- **Variate**

A variate is a particular/specific value of a variable.

- **Frequency**
  The number of times an entry (observation) is repeated.

- **Frequency Distribution**
  Frequency distribution is data containing frequency of each entry in tabular form

- **Ungrouped or Raw Data**
  Data in original form is called ungrouped data or raw data.

  e.g.,
  Following data represents marks obtained by 19 students in a class
  25, 6, 39, 44, 40, 29, 12, 42, 14, 16, 16, 20, 10, 33, 24, 25, 35, 30, 47.
  It is the raw data.

- **Array**
  An arrangement of ungrouped data in descending or ascending order is called an array.
  If we arrange the marks of 19 students in ascending order, we get the following array.
  6, 10, 12, 14, 16, 16, 20, 24, 25, 25, 29, 30, 33, 35, 39, 40, 42, 44, 47.

Frequency Distribution Table For Raw Data

Frequency distribution is tabular form of data containing frequency of each entry.
Frequency distributions are of 2 types :

(i) Discrete frequency distribution
(ii) Grouped or Continuous frequency distribution

Discrete Frequency Distribution

This distribution from the raw data is done by the method of tally marks. In the first column of the frequency table we write all possible values of the variable from the least to the greatest and in second column we write frequency of a particular value.

Grouped or Continuous Frequency Distribution

The discrete frequency distribution method of condensing the raw data is convenient only when the difference between the greatest and the smallest observations is not very large and the values in the raw data are largely repeating, if not so, then we use continuous frequency distribution.

Class-Interval

Class-interval is group in which the raw data is condensed.
Class limits are the limits by which each class if bounded. The value on the right side of a class is called upper limit and value on left side of a class is known as lower limit.

Measures of Central Tendency

A central value or an average value of a distribution is the value of variable which represents of the entire distribution, this representative value is the measure of central tendency.

There are five measures of central tendency.

(a) Mathematical average

(i) Mean or Arithmetic mean
(ii) Geometric mean
(iii) Harmonic mean

(b) Positional average

(i) Median
(ii) Mode

Arithmetic mean

(i) For ungrouped distribution

If $x_1, x_2, \ldots\ldots x_n$ are n values of variate x_i then their A.M. $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \ldots\ldots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n} \Rightarrow \quad \Sigma x_i = n\bar{x}$$

(ii) For ungrouped and grouped freq. distribution

If $x_1, x_2, \ldots. x_n$ are values of variate values of variate with frequencies $f_1, f_2, \ldots f_n$ respectively, then their A.M. is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots. + f_nx_n}{f_1 + f_2 + \ldots. + f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N},$$

where $N = \sum_{i=1}^{n} f_i$

Methods for calculating A.M.

(i) By direct formula

Simply use formula.

(ii) By Assumed mean

If the values of x_i are large, then the calculation is lengthy. In such cases we can take deviation of variate from an arbitrary point a (assumed mean).

Let $d_i = x_i - a$

$\therefore \bar{x} = a + \frac{\Sigma f_i d_i}{N}$, where a is assumed mean

(iii) By step deviation method

Short method can be used for finding the A.M. If each deviation d_i are divisible by a common number h(let)

Let $u_i = \frac{d_i}{h} = \frac{x_i - a}{h}$

$\therefore \quad \bar{x} = a + \left(\frac{\Sigma f_i u_i}{N}\right) h$

Q. **If for some $x \in R$, the frequency distribution of the marks obtained by 20 students in a test is :**

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x - 5$	$x^2 - 3x$	x

then the mean of the marks is

(A) 3.0 **(B) 2.5** **(C) 2.8** **(D) 3.2**

Sol. Given there are 20 students.

$Sf = (x+1)^2 + (2x - 5) + x^2 - 3x + x = 20$

$x^2 + 1 + 2x + 2x - 5 + x^2 - 3x + x = 20$

$\Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0$

$(x + 4)(x - 3) = 0 \Rightarrow x = 3$ or -4

$x = 3$ (rejecting negative value)

$\therefore f_2 = 16, f_3 = 1, f_4 = 0, f = 3$

$\because$ Mean $(\bar{x}) = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{32 + 3 + 0 + 21}{20} = \frac{56}{20} = 2.8$

Q. **Find the mean of the frequency distribution given below:**

x_i	5	12	20	24	30
f_i	10	14	10	5	7

Sol. Take assumed mean a = 15,

$u_i = x_i - 15$

$\therefore \quad \Sigma f_i u_i = 10(-10) + 14(-3) + 10(5) + 5(9) + 7(15) = 58$

$$\therefore \quad \bar{x} = a + \left(\frac{\Sigma f u_i}{N}\right) = 15 + \frac{58}{46} = 16.26$$

Q. **Suppose a given variable takes the value 0, 1, 2, ..., n with frequencies equal to the binomial coefficients $^nC_0$, $^nC_1$,.....,^nC_n respectively, then the mean of this distribution is**

(A) $\frac{n}{2}$ **(B)** $\frac{n+1}{2}$ **(C)** $\frac{n^2-1}{2n}$ **(D)** $\frac{n^2+1}{2n}$

Sol. **(A)**

$N = \Sigma f_i = [^nC_0 + {}^nC_1 + ... + {}^nC_n] = 2^n$

$\Sigma f_i x_i = [1.{}^nC_1 + 2.{}^nC_2 + ... + n\ {}^nC_n]$

$$= \sum_{r=1}^{n} r.{}^nC_r = n\sum_{r=1}^{n} {}^{n-1}C_{r-1} = n2^{n-1}$$

Thus $\bar{x} = \frac{1}{2^n}(n\ \ 2^{n-1}) = 0.5n$.

Q. **Following data represents the marks obtained by the students. Find the mean of marks.**

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Number of Students	10	22	25	18	21	4

Sol.

Marks	Class mark x	f	fx
0 - 10	5	10	50
10 - 20	15	22	330
20 - 30	25	25	625
30 - 40	35	18	630
40 - 50	45	21	945
50 - 60	55	4	220
		$\sum f = 100$	$\sum fx = 2800$

$$\therefore \quad \bar{x} = \frac{\sum fx}{\sum f} = \frac{2800}{100} = 28$$

(iv) Weighted arithmetic mean

If $w_1, w_2, \ldots, w_n$ be the weights of the values $x_1, x_2, \ldots, x_n$ taken in order then their weighted mean will be defined as

$$\text{Weighted mean} = \frac{w_1x_1 + w_2x_2 + \ldots + w_nx_n}{w_1 + \ldots + w_n} = \frac{\sum_{i=1}^{n} w_ix_i}{\sum_{i=1}^{n} w_i}$$

(v) Combined arithmetic mean

Suppose $\bar{x}_1$ and $\bar{x}_2$ be the means of two groups having n_1 and n_2 terms respectively then the combined arithmetic mean of their composite group can be calculated by :

$$\text{Combined arithmetic mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

For more than two groups, then

$$\text{Combined arithmetic mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3 + \ldots}{n_1 + n_2 + n_3 + \ldots}$$

Q. If we take weight of first n natural numbers equal to the corresponding number itself. What will be their weighted arithmetic mean ?

Sol. Weighted mean $= \dfrac{1.1 + 2.2 + 3.3 + \ldots + n.n}{1 + 2 + 3 + \ldots + n}$

$= \dfrac{1^2 + 2^2 + \ldots + n^2}{1 + 2 + \ldots + n}$

$= \dfrac{n(n+1)(2n+1)}{6\, n(n+1)/2} = (2n + 1)/3$

Q. **The weighted arithmetic mean of the first n natural numbers, the weight being the corresponding numbers, is**

(A) $\dfrac{n}{2}$ **(B)** $\dfrac{n+1}{2}$ **(C)** $\dfrac{2n+1}{3}$ **(D)** $\dfrac{n^2+1}{2n}$

Sol. **(C)**

First n natural numbers : 1, 2, 3, ..., n;

Corresponding weights are 1, 2, 3, ..., n respectively.

$\therefore$ Weighted arithmetic mean $= \dfrac{1\times1 + 2\times2 + \ldots\ldots + n\times n}{1 + 2 + \ldots\ldots + n}$

$= \dfrac{1^2 + 2^2 + \ldots\ldots + n^2}{1 + 2 + \ldots\ldots + n}$

$= \dfrac{n(n+1)(2n+1)}{\dfrac{6n(n+1)}{2}} = \dfrac{2n+1}{3}$

Q. **A factory is operating in two shifts, day and night, with 70 and 30 workers respectively. If per day mean wage of the day shift workers is ` 54 and per day mean wage of all the workers is 60, then per day mean wage of the night shift workers (in) is**

(A) 66 **(B) 69** **(C) 74** **(D) 75**

Sol. Let average wage of night shift workers be x.

Now, (70 × 54) + (30 × x) = 60 × 100

$\Rightarrow$ 3780 + 30x = 6000 $\Rightarrow$ 30 x = 2220 $\Rightarrow$ x = 74

Q. **The average salary of female employees in a firm was Rs. 4200 and that of male was Rs. 5200. Rs. 5000 was the mean salary of all the employees. The percentage of male is:**

(A) 80 **(B) 75** **(C) 90** **(D) 60**

Sol **(A)**

Let $x_1 = 5200$, $x_2 = 4200$, $\bar{x} = 5000$

Also, we know that $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$\Rightarrow 5000\,(n_1 + n_2) = 5200\,n_1 + 4200\,n_2$

$\Rightarrow \quad \frac{n_1}{n_2} = \frac{4}{1}$

$\therefore$ The percentage of male employees in the firm $= \frac{4}{4+1} \times 100 = 80\%$

and the percentage of female emplolyees in the firm $= \frac{1}{4+1} \times 100 = 20\%$

Properties of Arithmetic mean

- Algebraic sum of deviations of variate from their A.M. is always zero i.e. $\Sigma(x_i - \bar{x}) = 0$, $\Sigma f_i(x_i - \bar{x}) = 0$
- Sum of square of deviations of variate from their A.M. is always minimum i.e. $\Sigma(x_i - \bar{x})^2$ is minimum
- If $\bar{x}$ is the mean of variate x_i then
  If we add λ to each term then new mean= $\bar{x} + \lambda$

  If we multiply λ to each term then new mean = $\lambda\,\bar{x}$

  A.M. of $(ax_i + b) = a\bar{x} + b$
  (where λ, a, b are constant)
- Arithmetic mean does not depend on change of assumed mean i.e., it is not affected by any change in assumed mean.

Geometric mean

(i) **For ungrouped distribution :** Geometric mean G of n positive values x_1, x_2, ... x_n of variate is given by
$G = (x_1\ x_2\ .....\ x_n)^{1/n}$

$$\Rightarrow \quad G = \text{antilog}\left[\frac{1}{n}\sum_{i=1}^{n}\log x_i\right]$$

(ii) For frequency distribution: For n positive values $x_1, x_2, \ldots. x_n$ with corresponding frequencies $f_1, f_2, \ldots. f_n$ respectively, G.M will be given by

$$G = (x_i^{f_1} \bullet x_2^{f_2} \bullet \ldots.. \bullet x_n^{f_n})^{1/N}$$

$$\Rightarrow \quad G = \text{antilog}\left[\frac{1}{N}\sum_{i=1}^{n} f_i \log x_i\right]$$

Point to Remember!!!

For two groups having geometric means G_1 and G_2 and containing n_1 and n_2 positive values respectivel, then G is geometric mean of both groups then

$$G = (G_1^{n_1} \times G_2^{n_2})^{\frac{1}{n_1+n_2}}$$

$$\Rightarrow G = \text{antilog}\left[\frac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}\right]$$

Q. **The geometric mean of $1, 2, 2^2, \ldots, 2^n$ is**

(A) $\frac{n}{2}$ **(B)** $2^{\frac{n+1}{2}}$ **(C)** $2^{\frac{n}{2}}$ **(D)** $2^{\frac{n-1}{2}}$

Sol. **(C)**

$$\text{G.M.} = \left(1.2 \cdot 2^2 \ldots\ldots 2^n\right)^{\frac{1}{n+1}} = \left[2^{\frac{n(n+1)}{2}}\right]^{\frac{1}{n+1}} = 2^{n/2}$$

Harmonic mean

(i) For ungrouped distribution

If $x_1, x_2, \ldots, x_n$ are n non-zero values of a variate then their harmonic mean H is

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n}\frac{1}{x_i}}$$

(ii) For frequency distribution

If $f_1, f_2, \ldots. f_n$ are the frequencies of n non-zero values $x_1, x_2, \ldots, x_n$ respectively then their H.M.

$$H = \frac{N}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \ldots + \frac{f_n}{x_n}} = \frac{N}{\sum_{i=1}^{n} \frac{f_i}{x_i}}$$

Point to Remember!!!

If all entries are positive then A.M. ≥ G.M. ≥ H.M.

Q. **Find the H.M. of $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \ldots \frac{1}{17}$**

Sol. $$\text{H.M.} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots . + \frac{1}{x_n}} = \frac{16}{2 + 3 + \ldots . + 17} = \frac{2}{19}$$

Median

The median of data is the value of middle term of the observations when the values are written in descending or ascending order

Formulae of median

(i) For ungrouped distribution

If n is the number of observations in a series, then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ term}, & \text{when n is odd} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ terms}, & \text{when n is even} \end{cases}$$

(ii) For ungrouped frequency distribution

Make the cumulative frequency (c.f.) column and find the value of N (total number of elements) then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ term}, & \text{when n is odd} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ terms}, & \text{when n is even} \end{cases}$$

(iii) For grouped frequency distribution

Make c.f. column and find value of N/2, then median class is the class which contains value of c.f. is equal or just greater to N/2.

$$\therefore \quad \text{Median} = l + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

where l is lower limit of median class
and f is freq. of median class
and F is c.f. of the class preceeding median class
and h is Class interval of median class

Q. **Find the median of the average deposit balance of saving from given data.**

Average deposit balance (Rs.)	Number of deposit
Less than 100	26
100-200	68
200-300	145
300-400	242
400-500	188
500-600	65
600-700	16

Sol.

Average deposit balance (Rs.)	f	Cumulative frequency cf
Less than 100	26	26
100-200	68	94
200-300	145	239
300-400	242	481
400-500	188	669
500-600	65	734
600-700	16	750

$$\frac{N}{2} = \frac{750}{2} = 375$$

The frequency just greater than 375 is 481.

$\therefore$ Median class is 300 - 400.

$$\text{Median} = l + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

$$= 300 + \frac{375 - 239}{242} \times 100 \quad = 300 + 56.2 = 356.2$$

Q. **The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then y/x is equal to**

(A) 9/4 **(B) 7/2** **(C) 7/3** **(D) 8/3**

Sol. Given, mean = 42

$$\Rightarrow \frac{10 + 22 + 26 + 29 + 34 + x + 42 + 67 + 70 + y}{10} = 42$$

$$\Rightarrow \quad 300 + x + y = 420 \Rightarrow \quad x + y = 120 \ldots(i)$$

Also, median = 35 $\Rightarrow \frac{34 + x}{2} = 35 \Rightarrow x = 36$

From (i), we get $y = 120 - 36 = 84$

Now, $\frac{y}{x} = \frac{84}{36} = \frac{7}{3}$

Mode

In a frequency distribution, the value of the variate which has the maximum frequency is called mode.

(i) For ungrouped distribution

The value of that variate being repeated maximum number of times.

(ii) For ungrouped frequency distribution

The value of variate which having maximum frequency.

(iii) For grouped frequency distribution:

First we find the class which have maximum frequency, this is modal class

$\therefore$ $$\text{Mode} = l + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$$

where l — lower limit of modal class
f_0 — freq. of the modal class
f_1 — freq. of the class preceeding modal class
f_2 — freq. of the class succeeding modal class
h — class interval of modal class

Q. **If the mean of a set of observations $x_1, x_2, ..., x_{10}$ is 20, then the mean of $x_1 + 4, x_2 + 8, x_3 + 12, ..., x_{10} + 40$ is**
(A) 34 **(B) 32** **(C) 42** **(D) 38**

Sol. **(C)**

We have, $\bar{x} = \frac{\sum_{i=1}^{10} x_i}{10} = 20 \Rightarrow \sum_{i=1}^{10} x_i = 200$ (i)

Now, required mean $\frac{x_1 + 4 + x_2 + 8 + ... + x_{10} + 40}{10}$

$= \frac{\sum_{i=1}^{10} x_i + (4 + 8 + 12 + ... + 40)}{10} = \frac{200 + 4(1 + 2 + 3 + ... + 10)}{10}$

$= 20 + \frac{4 \times 10 \times 11}{2 \times 10} = 20 + 22 = 42$

Relation between Mean, Median and Mode

The relation is :
2 Mean + Mode = 3 Median

Point to Remember!!!

(i) Median always lies between mode and mean.

(ii) For a symmetric distribution the mode, mean and median are same.

Q. **In a moderately asymmetrical distribution, the mean and median are 36 and 34 respectively, find out the value of empirical mode.**

(A) 30 **(B) 32** **(C) 42** **(D) 22**

Sol. **(A)**

Mode = 3 Median − 2 Mean = 3 × 34 − 2 × 36 = 30

Q. **Given that a, b, c and d are real numbers following c< b< a< d. If mean and median of the distribution a, b, c, d are 5 and 6 respectively then find the value of a - 3d - 3c + b ?**

(A) -12 **(B) 16** **(C) 34** **(D) -23**

Sol. **(A)**

$\because \quad \frac{a+b+c+d}{4} = 5$

$\Rightarrow \quad a + b + c + d = 20$ ...(i)

and Arranging elements in descending order; d, a, b,c

$\therefore \quad \frac{a+b}{2} = 6$

$\Rightarrow \quad a + b = 12$ ...(ii)

From (i) − (ii), we have

$c + d = 8$ ...(iii)

Now, $-a + 3d + 3c - b$

$= 3(c + d) - (a + b)$

$= 3 \times 8 - 12$

$= 24 - 12 \quad = 12$

Measures of dispersion

The dispersion is the measurement of deviation of variate's values about their central/average value.

It provides an overview of scatteredness of different values from the average value.

There are 4 measures of dispersion.

(i) Range

(ii) Mean deviation

(iii) Variance and standard deviation

(iv) Quartile Deviation (Not in syllabus)

Q. **Which of the following is not a measure of dispersion:**
(A) Range (B) Mean deviation (C) Median (D) Standard deviation

Sol. **(C)**

There are 4 measures of dispersion.

(i) Range

(ii) Mean deviation

(iii) Variance and standard deviation

(iv) Quartile Deviation

(i) Range

Range of any distribution is the difference between the greatest and least values of variate.

In grouped distribution the range is the difference between upper limit of the maximum class and lower limit of the minimum class.

Know the facts



coefficient of range

$$= \frac{\text{difference of extreme values}}{\text{sum of extreme values}}$$

Q. **Find the range and coefficient of range of numbers 10, 8,12,11, 14, 9, 6**
(A) 16, 0.5 (B) 16, 0.2 (C) 8, 0.5 (D) 8, 0.4

Sol. **(D)**

As max value and minimum value of the distribution are 14 and 6 respectively, therefore

Range = 14 – 6 = 8

$$\text{Coefficient of range} = \frac{\text{difference of extreme values}}{\text{sum of extreme values}} = \frac{14-6}{14+6} = 0.4$$

(ii) Mean deviation (M.D.)

The mean of absolute value of deviations of variate from their statistical average(Mean, Median, Mode) is called mean deviation.

Suppose A is any statistical average of a distribution then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} |x_i - A|}{n}$$

[For ungrouped dist.]

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{N}$$

[For freq. dist.]

Point to Remember!!!

Mean deviation about the median is minimum.

Q. **If the values observed are 1, 2, 3, n each with frequency 1 and n is even, then the mean deviation from mean equals to**

(A) n **(B) n/2** **(C) n/4** **(D) None of these**

Sol. $\text{Mean} = \frac{\frac{n(n+1)}{2}}{n} = \frac{(n+1)}{2}$

Mean deviation from mean

$$= \frac{\left\{\sum_{i=1}^{n}\left|\frac{n+1}{2} - i\right|\right\}}{n} = \frac{1}{n} \times \left[\sum_{i=1}^{n/2}\left(\frac{n+1}{2} - i\right) + \sum_{i=\frac{n}{2}+1}^{n}\left(i - \frac{n+1}{2}\right)\right] \left[\because \frac{n+1}{2} - i < 0 \text{ for } i = \frac{n}{2}+1, \ldots, n\right]$$

$$\frac{1}{n} \times \left[\left\{\frac{n}{2} \times \frac{(n+1)}{2} - \frac{\frac{n}{2}\left(\frac{n}{2}+1\right)}{2}\right\} + \left\{\frac{n/2}{2}\left(\frac{3n}{2}+1\right) - \frac{n}{2}\frac{(n+1)}{2}\right\}\right]$$

$$= \frac{1}{n} \times \frac{n}{4}\left[\frac{3n}{2} + 1 - \frac{n}{2} - 1\right] = \frac{n}{4}$$

Q. **The coefficient of mean deviation about the median from the data (Weight (in kg) 54, 50, 40, 42, 51, 45, 47, 55, 57) is**

(A) 0.09000 **(B) 0.0956** **(C) 0.0056** **(D) 0.0946**

Sol. **(B)**

Median (m) = $\left(\frac{N+1}{2}\right)^{th}$ term

M.D. from Median = $\frac{\Sigma|x_i - m|}{N} = 4.78$

Coeff. of M.D. from the Median = $\frac{\text{M.D.}}{\text{Median}} = \frac{4.78}{50} = 0.0956$

Q. **For the following data, find the mean deviation about mean.**

x_i	3	9	17	23	17
f_i	8	10	12	9	5

Sol.

x_i	f_i	f_ix_i	$\lvert x_i - \bar{x}\rvert$	$\lvert x_i - \bar{x}\rvert$
3	8	24	12	96
9	10	90	6	60
17	12	204	2	24
23	9	207	8	72
27	5	135	12	60
	N = 44	$\Sigma f_ix_i = 660$		$\sum f_i\lvert x_i - \bar{x}\rvert = 312$

Mean = $\bar{x} = \frac{\Sigma f_ix_i}{N} = \frac{660}{44} = 15$

Mean deviation = $\frac{\Sigma f_i|x_i - \bar{x}|}{N} = \frac{312}{44} = 7.09$

Q. **For the following data, find mean deviation about the median.**

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of Students	6	8	14	16	4	2

Sol.

Class	f_i	cf	Mid value= x_i	$\lvert x_i - M \rvert$	$f_i \lvert x_i - M \rvert$
0 - 10	6	6	5	$\lvert 5 - 27.86 \rvert = 22.86$	137.16
10 - 20	8	14	15	$\lvert 15 - 27.86 \rvert = 12.86$	102.88
20 - 30	14	28	25	$\lvert 25 - 27.86 \rvert = 2.86$	40.04
30 - 40	16	44	35	$\lvert 35 - 27.86 \rvert = 7.14$	114.24
40 - 50	4	48	45	$\lvert 45 - 27.86 \rvert = 17.14$	68.56
50 - 60	2	50	55	$\lvert 55 - 27.86 \rvert = 27.14$	54.28
Total	$\Sigma f_i = 50$				517.16

$$\frac{N}{2} = \frac{50}{2} = 25$$

$\Rightarrow$ C = 14, f = 14, l = 20, h = 10

$$\text{Median } M = l + \frac{\frac{N}{2} - C}{f} \times h$$

$$= 20 + \frac{25 - 14}{14} \times 10 = 20 + \frac{11 \times 10}{14}$$

$$= 20 + 7.86 = 27.86$$

$\therefore$ Mean deviation about median $= \dfrac{\sum f_i \lvert x_i - M \rvert}{\sum f_i} = \dfrac{517.16}{50} = 10.34$

Formulae for variance

(i) For ungrouped distribution:

$$\text{Var}(X) = \sigma^2 = \frac{\Sigma (x_i - \bar{x})^2}{n}$$

$$\text{Var}(X) = \sigma^2 = \frac{\Sigma x_i^2}{n} - \bar{x}^2 = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2$$

$$\text{Var}(X) = \sigma^2 = \frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2, \quad \text{where } d_i = x_i - a$$

Proof If $x_1, x_2, \ldots, x_n$ are n values of a variable X, then

$$\sigma^2 = \text{Var}(X) = \frac{1}{n}\left\{\sum_{i=1}^{n}(x_i - \bar{x})^2\right\} \qquad \ldots(i)$$

$$\text{Now, Var}(X) = \frac{1}{n}\left\{\sum_{i=1}^{n}(x_i^2 - 2x_i\bar{x} + \bar{x}^2)\right\}$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}x_i^2 - \frac{1}{n}\sum_{i=1}^{n}2x_i\bar{x} + \frac{1}{n}\sum_{i=1}^{n}\bar{x}^2$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}x_i^2 - 2\bar{x}\left\{\frac{1}{n}\sum_{i=1}^{n}x_i\right\} + \frac{n\bar{x}^2}{n}$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}x_i^2 - 2\bar{x}^2 + \bar{x}^2 \qquad \left[\because \frac{1}{n}\sum_{i=1}^{n}x_i = \bar{x}\right]$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}x_i^2 - \bar{x}^2$$

$$\therefore \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}x_i^2 - \left\{\frac{1}{n}\sum_{i=1}^{n}x_i\right\}^2 \qquad \ldots(ii)$$

When the values of variable X are large, then for easing calculation we take deviations from an arbitrary point (A).

If $d_i = x_i - A$, $\quad i = 1, 2, \ldots, n$, then $\sum_{i=1}^{n}d_i = \sum_{i=1}^{n}(x_i - A) = \sum_{i=1}^{n}x_i - nA$

$$\Rightarrow \quad \frac{1}{n}\left\{\sum_{i=1}^{n}d_i\right\} = \frac{1}{n}\sum_{i=1}^{n}x_i - A$$

$$\Rightarrow \quad \frac{1}{n}\left\{\sum_{i=1}^{n}d_i\right\} = \bar{x} - A$$

$$\Rightarrow \quad \bar{d} = \bar{x} - A, \text{ where } \bar{d} = \frac{1}{n}\sum_{i=1}^{n}d_i$$

$$\therefore \quad \text{Var}(X) = \frac{1}{n}\left\{\sum_{i=1}^{n}(x_i - \bar{x})^2\right\}$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\left\{\sum_{i=1}^{n}\left(x_i - A + A - \bar{x}\right)^2\right\}$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}\left(d_i - \bar{d}\right)^2$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}\left(d_i^2 - 2d_i\bar{d} + \bar{d}\right)^2 = \frac{1}{n}\sum_{i=1}^{n}d_i^2 - \frac{1}{n}\sum_{i=1}^{n}2d_i\bar{d} + \frac{1}{n}\sum_{i=1}^{n}\bar{d}^2$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}d_i^2 - 2\left(\frac{1}{n}\sum_{i=1}^{n}d_i\right)\bar{d} + \frac{n\bar{d}^2}{n}$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}d_i^2 - 2\bar{d}^2 + \bar{d}^2$$

$$\Rightarrow \quad \text{Var}(X) = \frac{1}{n}\sum_{i=1}^{n}d_i^2 - \bar{d}^2$$

$$\Rightarrow \quad \text{Var}(X) = \frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2 \qquad \ldots(iii)$$

(ii) For frequency distribution:

$$\text{Var}(X) = \sigma^2 = \frac{\sum f_i\left(x_i - \bar{x}\right)^2}{N}$$

$$\text{Var}(X) = \sigma^2 = \frac{\Sigma f_i x_i^2}{n} - \bar{x}^2 = \frac{\Sigma f_i x_i^2}{n} - \left(\frac{\Sigma f_i x_i}{n}\right)^2$$

$$\text{Var}(X) = \sigma^2 = \frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2$$

$$\text{Var}(X) = \sigma^2 = h^2\left[\frac{\Sigma f_i u_i^2}{N} - \left(\frac{\Sigma f_i u_i}{N}\right)^2\right]$$

where $u_i = \frac{d_i}{h}$

Point to Remember!!!

Coefficient of S.D. $= \frac{\sigma}{\bar{x}}$

Coefficient of variation $= \frac{\sigma}{\bar{x}} \times 100$ (in percentage)

Q. **Calculate the variance of first n natural numbers**

Sol. $\sigma^2 = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2 = \frac{\Sigma n^2}{n} - \left(\frac{\Sigma n}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n} - \left\{\frac{n(n+1)}{2n}\right\}^2 = \frac{n^2-1}{12}$

Q. **If $\sum_{i=1}^{18}(x_i - 8) = 9$ and $\sum_{i=1}^{18}(x_i - 8)^2 = 45$, then the standard deviation of $x_1, x_2, \ldots x_{18}$ is:**

(A) 1.50 **(B) 5** **(C) 5.50** **(D) 3.50**

Sol. **(A)**

Let $(x_i - 8) = d_i$

$\therefore \quad \sigma = \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2} = \sqrt{\frac{45}{18} - \left(\frac{9}{18}\right)^2} = \sqrt{\frac{5}{2} - \frac{1}{4}} = \frac{3}{2}$

Q. **Find the coefficient of variation of first 100 natural numbers**

(A) $\sqrt{\frac{33}{101}}$ **(B)** $\sqrt{\frac{33}{101}} \times 100$ **(C)** $25\sqrt{3333}$ **(D) None of these**

Sol. **(B)**

For first n natural numbers.

$\text{Mean}(\bar{x}) = \frac{n+1}{2}, \text{S.D.}(\sigma) = \sqrt{\frac{n^2-1}{12}}$

$\therefore \quad$ Coefficient of variance $= \frac{\sigma}{\bar{x}} \times 100$

$$= \sqrt{\frac{n^2-1}{12}} \times \frac{1}{\left(\frac{n+1}{2}\right)} \times 100$$

$$= \sqrt{\frac{(n-1)}{3(n+1)}} \times 100$$

$$= \sqrt{\frac{99}{303}} \times 100$$

$$= \sqrt{\frac{33}{101}} \times 100$$

Q. **Find the standard deviation and variance for the following data**

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

Sol. Presenting the data in tabular form table, we get

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^2$
4	3	12	-10	100	300
8	5	40	-6	36	180
11	9	99	-3	9	81
17	5	85	3	9	45
20	4	80	6	36	144
24	3	72	10	100	300
32	1	32	18	324	324
	30	**420**			**1374**

$$N = 30, \sum_{i=1}^{7} f_i x_i = 420, \sum_{i-1}^{7} f_i (x_i - \bar{x})^2 = 1374$$

$$\text{Therefore } \bar{x} = \frac{\sum_{i=1}^{7} f_i x_i}{N} = \frac{1}{30} \times 420 = 14$$

$$\text{Hence variance } (\sigma) = \frac{1}{N} \sum_{i=1}^{7} f_i \ (x_i - \bar{x})^2$$

$$= \frac{1}{30} \times 1374 = 45.8$$

$$\text{Standard deviation } (\sigma) = \sqrt{45.8} = 6.77$$

Q. **Find the mean, standard deviation and variance for the following distribution:**

Class	10-20	20-30	30-40	40-50	50-60
Frequency	10	15	3	19	3

Sol. From the given data, we construct the following table

Class	Frequencey	Mid-Point	$x_i f_i$	$(x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^2$
10-20	10	15	150	324	3240
20-30	15	25	375	64	960
30-40	3	35	105	4	12
40-50	19	45	855	144	2736
50-60	3	55	165	484	1452
	50		**1650**		**8400**

Thus Mean = $\bar{x} = \frac{1}{N}\sum_{i=1}^{7} f_i x_i = \frac{1650}{50} = 33$

Variance = $(\sigma^2) = \frac{1}{N}\sum_{i=1}^{7} f_i (x_i - \bar{x})^2 = \frac{1}{50} \times 8400 = 168$

Standard deviation (σ) = $\sqrt{168} = 12.96$

Q. **Coefficient of variation of any two distributions are 70 and 60, and their standard deviations are 16 and 21 respectively. Find their arithmetic means.**

Sol. Given Coefficient of variation of 1st distribution = 60, $\sigma_1 = 21$
Coefficient of variation of 2nd distribution = 70, $\sigma_2 = 16$

Let $\bar{x}_1$ and $\bar{x}_2$ be the means of 1st and 2nd distribution, respectively.

Then C.V. (1st distribution) = $\frac{\sigma_1}{\bar{x}_1} \times 100$

i.e $60 = \frac{21}{\bar{x}_1} \times 100$

$\bar{x}_1 = 35$

and C.V (2nd distribution) = $\frac{\sigma_2}{\bar{x}_2} \times 100$

i.e $70 = \frac{16}{\bar{x}_2} \times 100$ or $\bar{x}_2 = \frac{16}{70} \times 100 = 22.86$

Q. **The variance of 20 observations is 5. Find the new variance if each observation is multiplied by 2.**

Sol. Let the observations be $x_1, x_2, ..., x_{20}$ and $\bar{x}$ be their mean.

Given that variance = 5 and n = 20. We know that

Variance = $\sigma^2 = \frac{1}{n}\sum_{i=1}^{20}\left(x_i - \bar{x}\right)^2$

$5 = \frac{1}{20}\sum_{i=1}^{20}\left(x_i - \bar{x}\right)^2$

or $\sum_{i=1}^{20}\left(x_i - \bar{x}\right)^2 = 100$ ..... (i)

If each observation is multiplied by 2, and the new resulting observations are y_1, then

$y_i = 2x_i$ i.e., $x_i = \frac{1}{2} y_i$

Therefore $\bar{y} = \frac{1}{n}\sum_{i=1}^{20} 2x_i = 2.\ \frac{1}{20}\sum_{i=1}^{20} x_1$

i.e. $\bar{y} = 2\bar{x}$ or $\bar{x} = \frac{1}{2}\bar{y}$

Substituting the values x_1 and $\bar{x}$ in (i) , we get

$\sum_{i=1}^{20}\left(\frac{1}{2}y_i - \frac{1}{2}\bar{y}\right)^2 = 100,$ i.e. , $\sum_{i=1}^{20}\left(y_i - \bar{y}\right)^2 = 400$

Thus, the variance of new observations = $\frac{1}{20} \times 400 = 20$

Q. **The mean and standard deviation of 100 observations were calculated as 40 and 5.1, respectively by a person who took by mistake 50 in place of 40 for one observation. The correct standard deviation is:**

(A) 5.5 **(B) 5** **(C) 6** **(D) None of these**

Sol. **(B)**

n = 100

Incorrect mean ($\bar{x}$) = 40

Incorrect standard deviation (σ) = 5.1

We know that $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

i.e., $40 = \frac{1}{100}\sum_{i=1}^{100} x_i$ or $\sum_{i=1}^{100} x_i = 4000$

i.e., Incorrect sum of observations = 4000

Thus, the correct sum of observations

= Incorrect sum – 50 + 40

= 4000 – 50 + 40 = 3990

Hence Correct mean = $\frac{\text{Correct sum}}{100} = \frac{3990}{100} = 39.9$

Also, Standard deviation $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2 - (\bar{x})^2}$

i.e., $5.1 = \sqrt{\frac{1}{100} \times \text{incorrect}\sum_{i=1}^{n} x_i^2 - (40)^2}$

or $100\ (26.01 + 1600) = \sum_{i=1}^{n} x_i^2$

Therefore, Incorrect $\sum_{i=1}^{n} x_i^2 = 162601$

Now, Correct $\sum_{i=1}^{n} x_i^2$ = Incorrect $\sum_{i=1}^{n} x_i^2 - (50)^2 + (40)^2$

= 162601 – 2500 + 1600 = 161701

Therefore, correct standard deviation

$= \sqrt{\frac{\text{correct}\sum x_1^2}{n} - (\text{Correctmean})^2}$

$= \sqrt{\frac{161701}{100} - (39.9)^2}$

$= \sqrt{1617.01 - 1592.01} = \sqrt{25} = 5$

Mean square deviation

The mean square deviation of a distrubution is the mean of the square of deviations of variate from assumed mean. It is denoted by S^2

Hence $S^2 = \dfrac{\Sigma(x_i - a)^2}{n} = \dfrac{\Sigma d_i^2}{n}$ [For ungrouped dist.]

$S^2 = \dfrac{\Sigma f_i (x_i - a)^2}{N} = \dfrac{\Sigma f_i d_i^2}{N}$ [For freq. dist.], where $d_i = (x_i - a)$

Q. **The mean square deviation of observations $x_1, x_2, ..., x_n$ about a point c is defined as $\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^2$. The mean square deviation about 2 and -2 are 10 and 18 respectively, then standard deviation of these observations is**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. **(C)**

$\frac{1}{n}\Sigma(x_i + 2)^2 = 18$ and $\frac{1}{n}\Sigma(x_i - 2)^2 = 10$

$\Rightarrow \Sigma(x_i + 2)^2 = 18n$ and $\Sigma(x_i - 2)^2 = 10n$

$\Rightarrow \Sigma(x_i + 2)^2 + \Sigma(x_i - 2)^2 = 28\,n$ and $\Sigma(x_i + 2)^2 - \Sigma(x_i - 2)^2 = 8\,n$

$\Rightarrow 2\Sigma x_i^2 + 8n = 28\,n$ and $8\Sigma x_i = 8n$

$\Rightarrow \Sigma x_i^2 = 10\,n$ and $\Sigma x_i = n$

$\Rightarrow \dfrac{\Sigma x_i^2}{n} = 10$ and $\dfrac{\Sigma x_i}{n} = 1$

$\therefore \sigma = \sqrt{\dfrac{\Sigma x_i^2}{n} - \left(\dfrac{\Sigma x_i}{n}\right)^2} = \sqrt{10 - (1)^2} = 3$

Relation between mean square deviation and variance

$\sigma^2 = \dfrac{\Sigma f_i d_i^2}{N} - \left(\dfrac{\Sigma f_i d_i}{N}\right)^2$

$\Rightarrow \sigma^2 = s^2 - d^2$, where $d = \bar{x} - a = \dfrac{\Sigma f_i d_i}{N}$

$\Rightarrow s^2 = \sigma^2 + d^2$

$\Rightarrow s^2 \geq \sigma^2$

Hence, the minimum value of mean square deviation is variance.

Properties of Variance

(i) $\text{Var.}(x_i + \lambda) = \text{Var.}(x_i)$

(ii) $\text{Var.}(\lambda x_i) = \lambda^2.\text{Var}(x_i)$

(iii) $\text{Var.}(ax_i + b) = a^2.\text{Var}(x_i)$

where λ, a, b, are constant

Point to Remember!!!

- If means of two series containing n_1, n_2 terms are $\bar{x}_1$ and $\bar{x}_2$ and their variances are σ_1^2 and σ_2^2 respectively and their combined mean is $\bar{x}$ then the variance of their combined series is given by following formula

$$\sigma^2 = \frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{\left(n_1 + n_2\right)}$$

where $d_1 = \bar{x}_1 - \bar{x}, d_2 = \bar{x}_2 - \bar{x}$

i.e.,,

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{\left(n_1 + n_2\right)^2}\left(\bar{x}_1 - \bar{x}_2\right)^2$$

Q. **The standard deviation and mean of marks obtained by 50 students of a class in three subjects Mathematics, physics and chemistry are as follows :**

Subject	Mathematics	Physics	Chemistry
Mean	42	32	40.9
Standard deviation	12	15	20

Which of these three subjects shows the lowest variability in marks and which show the highest ?

Sol. Here, n = 50

For Mathematics, Coefficient of variation (CV) $= \frac{\sigma}{\bar{x}} \times 100$

$$= \frac{12}{42} \times 100 = \frac{2}{7} \times 100 = \frac{200}{7}$$

$= 28.57$ ...(i)

For Physics, $CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{15}{32} \times 100$

$= \frac{1500}{32} = 46.87$ ...(ii)

For Chemistry, $CV = \frac{\sigma}{\bar{x}} \times 100$

$\frac{20}{40.9} \times 100 = \frac{2000}{40.9} = 48.89$ ...(iii)

From Equations: (i), (ii) and (iii), we have
CV of chemistry > CV of Physics > CV of Mathematics
∴ Mathematics shows the least variability and Chemistry shows the highest variability.

Q. **The mean and standard deviation of marks obtained by 50 students of a class in three subjects, Mathematics, Physics and Chemistry are given below**

Subject	Mathematics	Physics	Chemistry
Mean	42	32	40.9
Standard deviation	12	15	20

Which of these three subjects shows the highest variability in marks and which shows the lowest?

(A) Chemistry, Mathematics **(B) Mathematics, Physics**
(C) Chemistry, Physics **(D) None of these**

Sol. **(A)**

Here, n = 50
For Mathematics, $C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{12}{42} \times 100 = 28.57$ ...(i)

For Physics, $C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{15}{32} \times 100 = 46.88$ ...(ii)

For Chemistry, $C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{20}{40.9} \times 100 = 48.90$ ...(iii)

From (i), (ii) and (iii), we have
C.V. of Chemistry > C.V. of Physics > C.V. of Mathematics
∴ Chemistry shows the highest variability and Mathematics shows the least variability.